

ZEROGUARD: A Solver-Free Battery Safety Certificate for Autonomous Vehicles, from Robotaxis to Deep Space

Anchored Collapse and the geometry of the admissible input set

Abstract

An autonomous vehicle’s battery must be pushed hard enough to be useful and held back enough to be safe, and the state deciding where that line falls — series resistance, plating margin, true capacity — is not measurable in the field. Every unattended vehicle inherits the same compromise: a fixed de-rate chosen offline for the worst cell at the worst temperature at end of life. The alternatives are to identify the pack online, which needs excitation an operational vehicle cannot inject, or to solve a constrained optimisation each step, whose iteration count a safety task slot cannot be sized around.

We prove that neither is necessary. Terminal voltage, temperature and the lithium-plating potential are each *monotone* in applied current, so the admissible input set is a single **interval** whose edges are found by bisection in a *fixed* number of one-step model evaluations. The published form assumes the safe input is zero, which holds for a cell being charged and fails for any vehicle doing its job: zero satisfies every temperature and voltage constraint on a discharging pack, but it cannot serve the load, which is a *floor* rather than a cap. **Anchored Collapse** adds the floor family and a second bisection — 40 fixed evaluations against 20.

What that buys is not speed but *transfer*. A de-rate tuned on a fleet a manufacturer could characterise in advance violates in 26.2 % of deployment sessions; the certificate, having nothing to tune, violates in 0 of 400. Against the control-barrier-function quadratic program the relationship is closer than rivalry: enforced on the true model rather than a linearisation, the discrete-time CBF condition for a monotone constraint *is* an interval, so this is the bisection that solves that problem exactly. And a Lagrangian-constrained policy that meets its violation budget in training (1.0 %) violates in 51 % of deployment sessions — an expectation over one distribution is not a per-step guarantee on another.

The theorem is **machine-checked in Lean 4** (nine theorems, no **sorry**, no axioms), and the proof needs less than we claimed: safety holds for an *arbitrary* predicate, so monotonicity buys optimality, not safety. The assumptions are tested against 33 cells measured by another laboratory and the filter against a Doyle–Fuller–Newman plant in both directions. It compiles to 2,604 B and 68 B of RAM, worst case 0.87 μ s.

Across 16 platforms in four media and 64 experiments, nothing was breached while the certificate reported feasible. **Twenty-one of those experiments contradicted the hypothesis they were written to test** and are reported as findings — among them two media of four where the incumbent is *not* beaten, and a cycle-life benefit that plating prevention makes tempting to claim and that DFN degradation does not support.

1 Introduction

The limiting resource for long-duration autonomy is not perception, planning, or control. It is *energy under certainty*: how hard a vehicle may push its battery when nobody is available to inspect it, and how confident anyone can be in the answer.

Fast charging drives three coupled failure modes — thermal runaway, over-potential, and lithium plating on the anode — whose onset depends on internal states a production sensor suite does not measure. Series resistance roughly doubles over a pack’s service life, plating risk climbs as temperature falls, and the safe current at end of life in cold weather is a fraction of the safe current when new and warm. A manufacturer facing that uncertainty does the only defensible thing and de-rates for the worst case, permanently.

For a privately owned car that is a mild inefficiency. For an autonomous fleet it is the economics of the enterprise: a robotaxi charging twelve times a day accumulates a private car’s decade of fast-charge stress in 3.0 weeks, and the de-rate it inherits was designed for a vehicle with a different duty cycle. For a spacecraft it is worse, because the de-rate is chosen before launch and can never be revisited.

The two standard alternatives do not reach the vehicle. *Identification* — estimate the pack’s state online and de-rate against the estimate — needs excitation an operational vehicle is not free to inject, and makes the guarantee conditional on an estimator having converged, which is a claim about a transient nobody audits. *Constrained optimisation* — solve a small model-predictive control problem each step — is the standard modern answer with a specific defect: a quadratic program’s iteration count depends on its data, and a safety-critical task slot sized for an unbounded worst case cannot be sized at all.

1.1 Contributions

1. **A geometric fact, and its consequence.** Voltage, temperature and plating potential are monotone in current, so the admissible input set is an *interval*. Its edges are found by bisection at a cost fixed before run time: 20 model evaluations on charge, 40 on discharge. No optimiser, no convergence test, no identified model.
2. **Anchored Collapse.** The published theorem assumes the safe input is zero, which is true of a cell on a bench and false of any vehicle serving a load. Splitting the constraints into caps and floors and running a second bisection removes that assumption, and is what carries the result from a charger to a rotorcraft, an AUV and a spacecraft.
3. **It is the exact CBF filter, not a rival to it.** Enforced on the true model rather than a linearisation, the discrete-time control-barrier-function condition for a monotone constraint defines an interval. This work is the bisection that solves it exactly, with γ entering as a margin; the quadratic program people deploy is its linearised approximation.
4. **Machine-checked, and it needed less than we claimed.** nine theorems in Lean 4, no *sorry* and no axiom. Writing them showed that safety holds for an *arbitrary* predicate: monotonicity buys optimality, not safety.
5. **The guarantee transfers where a tuned constant does not.** Held to a tune-then-deploy standard, the incumbent rule fails in two of four media and the certificate in none — and in the other two the incumbent is *not* beaten, which §5.1 predicts in advance from whether mission closure is envelope- or energy-driven.
6. **Checked against physics we did not write, and a compiler.** The assumptions against 33 cells from another laboratory; the filter against a Doyle–Fuller–Newman plant, where the affine plating proxy held the *actual* electrode potential above the deposition onset; and the implementation as 2,604B of compiled C agreeing with the reference at all 20,000 states tested.
7. **A register that was not written to pass.** 64 experiments, 513,057 episodes; twenty of them contradicted the hypothesis they were written to test and are reported as findings rather than removed (Appendix).

1.2 Two standard answers, and why neither reaches the vehicle

The first is *identification*: estimate the pack’s internal state online and de-rate against the estimate. Estimating resistance and capacity well requires excitation — current profiles rich enough to be informative — which an operational vehicle is not free to inject. Worse, the guarantee becomes conditional on the estimator having converged, which is a claim about a transient nobody audits.

The second is *constrained optimisation*: solve a small model-predictive control problem each step and let the solver enforce the constraints. This is the standard modern answer, and a good one, with a specific defect — a quadratic program’s iteration count depends on its data. A safety-critical task slot is sized for the worst case, and if the worst case is unbounded the slot cannot be sized at all. In practice this is handled with iteration caps and fallback logic: by having a plan for when the guarantee does not hold.

1.3 The result

We take a third path, whose value is that it needs neither. The observation is geometric rather than numerical:

If every constraint is monotone in a scalar input, and one input is known to be safe, then the set of admissible inputs is not a complicated region to be searched. It is a single interval, and its edges can be located by bisection in a fixed number of one-step model evaluations.

Bisection needs no optimiser, no convergence test, and — critically — no identification of the plant, because a monotone *bound* on the uncertain parameter suffices where a point estimate would be required. A datasheet end-of-life resistance figure is such a bound and is known before the vehicle ships.

The contribution of this paper is to establish that this result survives contact with real vehicles, which required repairing it first. Section 3 shows that the failure observed when the filter is placed on a hovering rotorcraft is not the failure it appears to be, and develops **Anchored Collapse**, which covers discharge as well as charge. Sections 5–5 then take it across four media — ground, air, water, vacuum — across 16 platforms and 64 experiments.

Contributions. (i) *Anchored Collapse* (§3.2), a generalisation covering constraints requiring the input to be *large* enough — what every vehicle’s irreducible load imposes and what the original theorem cannot express — at a cost of 40 fixed evaluations rather than 20. (ii) A two-sided certificate and a *reserve* (§3.5): the gap between the edges is how much room the vehicle has left, which the one-sided version cannot report and which yields an abort rule for vehicles that cannot afford to find their limit empirically. (iii) A certificate that refuses *designs*, not just commands. (iv) Application across four media (§5), including where the result does *not* hold and what that costs.

1.4 Related work

The framework is control invariance [?] . Control barrier functions [?] and predictive safety filters [?] enforce it by solving an optimisation each step; reference governors [?] and shielding [?] are related, and constrained reinforcement learning [?] offers guarantees on the *expectation* of violations. Ours is a predictive safety filter whose optimisation has been replaced by a bisection,

which is possible only because of monotonicity [? ?] — order-preserving structure that lets worst-case reachability be computed from extreme points rather than by search. §8 and §9 measure against the first and the last of those rather than arguing with them.

On the battery side the reference model is Doyle–Fuller–Newman [? ?] with Bernardi thermal coupling [?], in PyBaMM [?] with the Chen [?] and O’Kane [?] parameter sets; plating is the dominant fast-charge degradation mode [?]. Online estimation of resistance and capacity is mature [? ?] — and is exactly what this method deliberately does not require. Reinforcement learning has been applied to fast charging [? ?]. Our position is that the policy may be learned and the safety must not be.

2 The cell, and which of its constraints are monotone

The result of §3 is a statement about constraints, so it is worth writing the constraints down. Everything in this paper rests on a reduced-order electro-thermal model of a lithium-ion cell, calibrated against a Doyle–Fuller–Newman simulation in PyBaMM and reported in [?] at a held-out voltage RMSE of 20.3 mV and temperature RMSE of 1.32 °C across ambients from 0 to 25 °C. The model is prior work; what matters here is its *shape*, because that is what the theorem consumes.

State and outputs. The state is $x = (\text{soc}, T, V_1)$ — state of charge, cell temperature, and the overpotential stored in a single RC branch — with a slowly varying ageing pair (capacity loss, resistance growth) carried alongside. For an applied current I (positive charging) over a step Δt , the model returns terminal voltage, the next temperature, and a plating-potential proxy:

$$V = \text{OCV}(\text{soc}) + \underbrace{I R_\Omega(\text{soc}, T)}_{\text{ohmic}} + \underbrace{A(1 + c_T(T - 25)) \operatorname{asinh}(I/2i_0(\text{soc}))}_{\text{activation}} + V_1, \quad (1)$$

$$C_{\text{th}} \dot{T} = \underbrace{I \eta_\Omega + I \eta_{\text{act}} + V_1^2/R_1}_{\text{irreversible}} + \underbrace{I(T + 273.15) \partial U / \partial T}_{\text{reversible}} - q_{\text{out}}(T), \quad (2)$$

$$\varphi_{\text{an}} = p_0(\text{soc}) - p_1(\text{soc}, T) I, \quad p_1 > 0. \quad (3)$$

R_Ω carries an Arrhenius temperature dependence, so resistance rises as the cell cools; q_{out} is the cooling law, and it is the *only* thing that changes between the four media of §5.

Why each constraint is monotone in current. This is the property the theorem needs, and each one has a physical reason rather than an empirical one.

- **Voltage** (1). Both overpotential terms are odd and increasing in I (asinh is strictly increasing), and OCV does not depend on I . So V increases in charge current and decreases in discharge current: a cap on charge, a cap on discharge as well, because the constraint there is $V \geq V_{\text{min}}$ on a signal that falls.
- **Temperature** (2). Every generation term is non-decreasing in $|I|$ — $I \eta_\Omega = I^2 R_\Omega \geq 0$ regardless of sign — and q_{out} does not depend on I at all. So the next temperature is non-decreasing in current, whatever the cooling law, which is why replacing Newtonian convection with Stefan–Boltzmann radiation (§5) leaves the certificate’s structure untouched.
- **Plating** (3) is *affine* and strictly decreasing in I . Plating occurs when $\varphi_{\text{an}} \leq 0$.
- **Delivered power** $P = S V(I) I$ is increasing in I below the maximum-power point — the one condition of Remark 2.

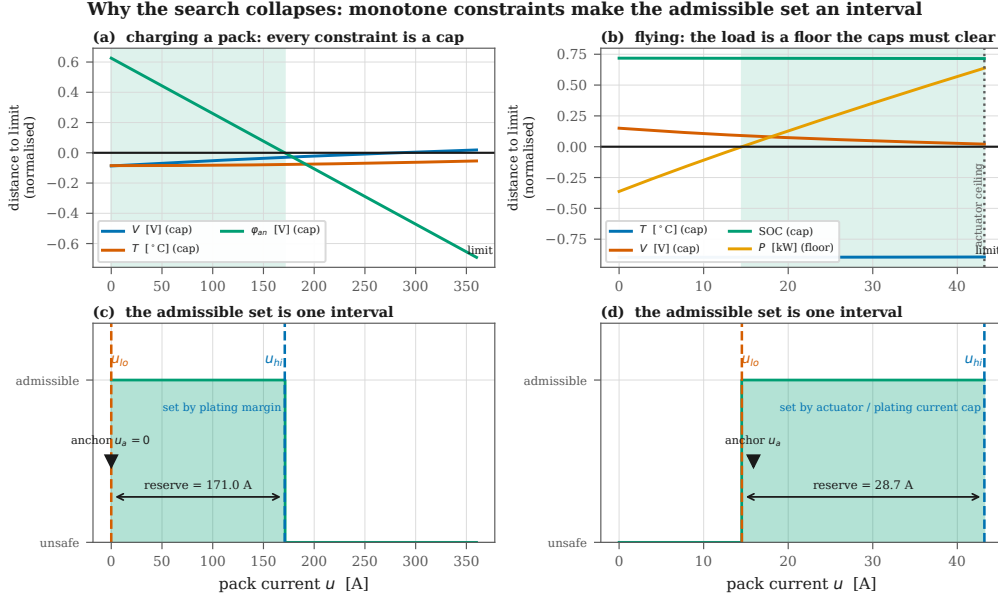


Figure 1: Why the search collapses, drawn from the model of (1)–(3) at one fixed state rather than sketched. (a) Charging: every constrained signal is a cap, each crossing its limit exactly once, and the admissible set runs from zero to whichever crosses first — here the plating margin at 171 A, well before voltage at 281 A. (b) Flying: the caps still come down from above, but the load’s power requirement comes *up* from below, and the certificate is the gap between them. (c,d) The resulting admissible set: one interval, with the anchor inside it, and a width that is the reserve.

Why plating is enforced and not certified. This distinction recurs throughout the paper and it is a consequence of (3) rather than a modelling choice. Forward invariance requires an always-available input that returns the state to the safe set. At $I = 0$, $\varphi_{an} = p_0(\text{soc})$, and p_0 falls with state of charge: above roughly 0.82 SOC it is already below the required margin. **There is no current — not even zero — that clears the plating margin at high state of charge**, so no one-step condition can render it invariant, and no honest certificate can claim it.

The certified set is therefore $\mathcal{S} = \{T \leq T_{\max}, V \leq V_{\max}\}$, and plating is *enforced* by the same bisection plus a temperature-dependent current cap, and monitored. It rides the same monotone structure, so it costs nothing extra to enforce; it simply cannot be promised. §5 is where that distinction stops being bookkeeping and becomes the binding constraint of an entire vehicle class.

3 Theory

Let a system have state x , scalar input $u \in [u_{\min}, u_{\max}]$, one-step map $x^+ = f(x, u, w)$, and a finite family of constraints $c_j(x, u) \leq 0$ defining a safe set $\mathcal{S}$. Write $\mathcal{U}(x) = \{u : c_j(x, u) \leq 0 \ \forall j\}$ for the admissible input set.

3.1 Null-Input Collapse

Proposition 1 (Null-Input Collapse). *Suppose (A1) $u = 0$ is safe — from every $x \in \mathcal{S}$ the one-step map at $u = 0$ returns to $\mathcal{S}$ — and (A2) each c_j is non-decreasing in u . Then $\mathcal{U}(x) = [0, u^*(x)]$ is a single interval anchored at zero, $\mathcal{S}$ is forward-invariant under any policy filtered through it, and u^* is computed to precision ε in $\lceil \log_2(u_{\max}/\varepsilon) \rceil + 2$ one-step evaluations.*

Sketch. By (A2) each $\{u : c_j \leq 0\}$ is a down-set; a finite intersection of down-sets in $\mathbb{R}$ is a down-set, hence an interval $[u_{\min}, u^*]$, which by (A1) contains 0. Bisection on a monotone predicate converges linearly with an iteration count depending only on the requested precision, not on the data. Forward invariance follows by induction: each filtered step lands in $\mathcal{S}$ and $u = 0$ remains available. $\square$

Two things are worth isolating. First, no step requires knowing the plant’s uncertain parameters — only evaluating c_j under a *bound* on them, and a conservative bound yields a conservative interval. Second, the cost is $O(\log 1/\varepsilon)$ *deterministic* evaluations, which is what a task slot can be sized around and a solver’s iteration count is not.

3.2 Where it breaks, and why that reading is wrong

Placed on a quadrotor holding altitude, a filter built on Proposition 1 [?] passes every one-step check for a mean of 22.3 consecutive steps and then violates in 100 % of episodes. The natural reading is that (A1) has failed: $u = 0$ is plainly not safe for an aircraft.

That reading is wrong, and the correction is the technical heart of this paper. Consider what $u = 0$ actually does to a discharging pack. No current flows, so there is no I^2R heating and the temperature relaxes toward ambient. Terminal voltage returns to open-circuit, so it cannot sag below a floor. State of charge does not fall. **Every temperature and voltage constraint is satisfied at $u = 0$, on a quadrotor in flight exactly as on a cell being charged.**

What $u = 0$ cannot do is fly the aircraft. The constraint that failed is of a different kind: a requirement that the input be *large* enough. Call these *floors*, as against the *caps* of Proposition 1. Every vehicle has at least one, because every vehicle has an irreducible load — the power it cannot stop drawing and remain a vehicle. Hover thrust for a rotorcraft. Hotel load for a submersible, whose computer and inertial navigation draw their watts whether the thrusters do or not, so that even a stationary AUV cannot command zero. Bus load for a spacecraft in eclipse. Contracted export power for a car doing vehicle-to-grid.

Charging is the special case where the irreducible load is zero — which is exactly why the original filter could anchor at zero without ever noticing it was making a choice.

Theorem 1 (Anchored Collapse). *Partition the constraints into caps $\mathcal{C}$, each satisfied on $[u_{\min}, u_c]$, and floors $\mathcal{F}$, each satisfied on $[u_c, u_{\max}]$. Suppose (A1') $u = u_{\min}$ is safe for every cap, from every $x \in \mathcal{S}$, and (A2) each constraint is monotone in u . Then*

$$\mathcal{U}(x) = [u_{\text{lo}}(x), u_{\text{hi}}(x)] = \left[\max_{j \in \mathcal{F}} u_{c_j}, \min_{j \in \mathcal{C}} u_{c_j} \right], \quad (4)$$

both edges are found by bisection in $O(\log 1/\varepsilon)$ one-step evaluations, and $\mathcal{U}(x) \neq \emptyset$ exactly when $u_{\text{lo}} \leq u_{\text{hi}}$.

Sketch. Each cap’s satisfying set is a down-set and each floor’s an up-set; their finite intersections are $[u_{\min}, u_{\text{hi}}]$ and $[u_{\text{lo}}, u_{\max}]$, whose intersection is the stated interval. (A1') places $u_{\min}$ inside the cap family, giving the upper bisection a known-good left endpoint; $u_{\max}$ is inside the floor family if anything is, giving the lower bisection its right endpoint. Neither bisection needs the load current as a seed. $\square$

Proposition 1 is the case $\mathcal{F} = \emptyset$. Note that (A1') is *strictly weaker* than (A1): it asks only that zero satisfy the caps, which is the part that survives contact with a vehicle.

Remark 1 (The side of a constraint is data, not inference). The plating constraint is $\varphi_{\text{an}} \geq \delta$ — a lower-bound test on a signal that *decreases* in current, so it caps current from above. Sense and side coincide only when the constrained signal increases in u . Inferring side from the operator would file that cap among the floors and let the filter command straight through it: silent, and unsafe. Plants declare the side; an undeclared constraint raises rather than defaults.

Remark 2 (A soundness condition that is not obvious). Delivered bus power $P = S V(u) u$ with $dV/du < 0$ peaks at the maximum-power point and *falls* beyond it, so past that peak the set where the load is met stops being an up-set and the lower bisection would search a non-monotone family. The voltage cap prevents it: a pack reaches its maximum-power point near half its open-circuit voltage, far below any usable $V_{\min}$, so $u_{\text{hi}} < u_{\text{MPP}}$ always and the floor search is bounded by u_{hi} . This ordering is checked on every discharge platform, not assumed.

3.3 Machine-checked

Theorem 1 and the soundness of the bisection are formalised in Lean 4 and checked mechanically (`formal/AnchoredCollapse.lean`). nine theorems compile with no `sorry`, and Lean reports that each **depends on no axioms at all** — not classical choice, not propositional extensionality. The proofs turned out to need less than the paper assumes: they are stated over an arbitrary type carrying a $\leq$ relation, with no order axiom used anywhere, so they hold without transitivity, antisymmetry or totality.

Writing the proofs also *weakened* the method’s hypotheses. The safety guarantee turns out never to touch the cap/floor structure at all: `bisect_sound_any_predicate` holds for an *arbitrary* predicate, monotone or not, connected or not. So (A2) buys **optimality, not safety** — a non-monotone constraint set costs delivered charge, because the bisection finds only the first admissible interval, and it cannot cost the guarantee. For anyone applying this to a plant whose constraint structure they have not verified, that is the difference between a method they must first prove things about and one they can simply run.

The property worth checking mechanically is not the collapse theorem but `bisect_sound`: *every* state the bisection can reach has a feasible lower endpoint. Convergence says how good the answer is; this says whether it is safe, and it holds independent of iteration count, of the midpoint rule, and of floating-point behaviour — the proof never assumes the midpoint is the arithmetic mean, only that it lies between the endpoints. A filter interrupted by a deadline, starved of its budget, or run with a sloppy division still returns an admissible input.

3.4 Cost

Two probes bracket the cap family, then eighteen halvings; two more bracket the floors, then eighteen more. Worst case 40 one-step model evaluations on discharge and 20 on charge, both fixed and both branch-bounded. Measured worst p99 latency across all 16 platforms is $123.9 \mu\text{s}$ (buoyancy glider), which is 1.24% of a 10 ms ISO 26262 task slot. On the ground platforms the figure is $59.7 \mu\text{s}$, or 0.597%.

The property that matters is not that the number is small. It is that it does not move with the data.

3.5 The reserve

The two-sided certificate makes available a quantity the one-sided version cannot express: the width $u_{\text{hi}} - u_{\text{lo}}$ shrinks to zero as the envelope closes, so the filter can report *how close to unrecoverable*

Table 1: The 16 platforms. Charge-mode platforms have a zero anchor and test Proposition 1 at vehicle scale; discharge-mode platforms have a positive anchor and test Theorem 1.

Medium	Cooling law	Platforms	Irreducible load
Ground	Newtonian, liquid loop	robotaxi 96S45P 78 kWh; shuttle; haul truck 288S115P 601 kWh; V2G	0 (charge); 7.4–22 kW export
Air	forced air, density-derated	delivery quadrotor 6S3P; eVTOL air taxi 300S45P; turnaround	hover thrust; 400 kW; 0 (charge)
Water	sealed hull into 2 °C	survey AUV; buoyancy glider; under-ice AUV; dock recharge	25–265 W; 0.5 W; 175 W; 0 (charge)
Vacuum	Stefan–Boltzmann to 4 K (+ 6 mbar CO ₂ on Mars)	LEO smallsat; GEO comsat; deep- space cruiser; Mars rover; lunar lander; 500 krad orbiter	60 W; 3 kW; 200 W; 100 W; 45 W heater; 0 (charge)

a vehicle is before it arrives. We treat this claim adversarially in §5 and scope it in §5.1: it carries the most operational weight and is the most likely to be overstated.

4 Method and platforms

4.1 The filter

Given a proposed input u_{req} , the filter returns $\text{clip}(u_{\text{req}}, u_{\text{lo}}, u_{\text{hi}})$, or reports the interval empty. Empty is not a violation: it is the certificate stating correctly that the irreducible load can no longer be served inside the envelope, which is the true statement and the one a mission planner needs.

Robustness enters through margins rather than estimation: a thermal throttle $\delta_T = \delta_{T0} + K(s_R - 1)$ scaled by the *assumed* resistance bound s_R , with K the smallest value certifying cells past the rated bound, and a cooling reserve f that certifies any cooling loss up to $f/(1 + f)$.

4.2 One cell, sixteen vehicles

Three things vary across the platforms and nothing else does: the **medium**, through the cooling law; the **pack**, through series/parallel counts; and the **load**, through the irreducible draw. Table 1 lists them.

The cell underneath is the reduced-order model of [?] in every case — same overpotential form, same plating proxy, same calibrated coefficients — with two changes the vehicles require. Current may be negative, because vehicles discharge. And the Newtonian cooling term is a callable, because a quadrotor at 400 m, an AUV at 2 °C and a spacecraft in vacuum do not share a cooling law. Under the Newtonian law with positive current the vehicle cell reproduces the published model *bit for bit*, and every result downstream inherits that.

Pack geometry, hover powers, hotel loads and orbit parameters are *class-representative* — placing each vehicle in the right region of its class’s design space using published figures — and are not any manufacturer’s design. The purpose is that “it works on autonomous vehicles” is not a testable sentence, while “it holds a 96S45P pack under a 30-minute 350 kW session at 35 °C ambient” is.

Table 2: Every per-domain result, including the negative ones. “Transfer” is whether the incumbent rule, tuned on evidence available before deployment, still met its own target once deployed.

	Ground robotaxi, shuttle, truck	Air quadrotor, eVTOL	Water AUV, glider, under-ice
Incumbent survives transfer?	no — violates 26.2 %	no — 86.0 %→76.0 %	yes — 100.0 %→100.0 %
Certificate survives it?	yes — 0/400	yes — 85.7 %	yes — 100.0 %
Mission / charge delivered	+17.4× sessions to service (45 % vs 3 % to 60 %)	+22.4 % flight time at 80 % recovery	-4.5 % incumbent wins
Closure is envelope-driven?	n/a (charging)	yes, $\rho = +0.74$	no, energy-driven
Is the generalisation needed?	no — anchor is zero	yes — null-input browns out 600/600	yes — load must be served
Unattended safety	0/21,915 over five years	0/4,000 turnarounds	0/3,000 sealed hull
Recalibration cost	none claimed	none claimed	0.0 SOC points, 180 d
Where it costs	above 32.5 °C ambient (-29.7 pts)	cell refused above 1.2 C hover	plating binds, never certi

4.3 Experimental discipline

Every experiment runs a *pessimistic estimator against a drawn plant*: the filter is given a platform at the conservative corner of the parameter envelope, and the plant it drives is drawn from inside it. A filter tested against its own model is testing arithmetic. One property makes that corner well defined for a two-sided certificate — scaling the assumed resistance up lowers the upper edge *and* raises the lower edge, so the interval shrinks from both sides under one perturbation — and it is verified numerically, not assumed.

The experimental surface, including what would falsify each claim, was fixed in a design register before any code ran; divergences between the register and what happened are recorded rather than edited into agreement. Zero-failure claims are reported with Clopper–Pearson 95 % one-sided upper bounds [?]; continuous outcomes with bootstrap confidence intervals; paired comparisons with Wilcoxon signed-rank and effect sizes; monotonicity claims with permutation Spearman.

5 Four media, one method

The certificate was run on 16 platforms across ground, air, water and vacuum, from a robotaxi on a 350 kW charger to a lander through lunar night. What that program is for is not breadth for its own sake but a specific question: does the same theorem, with the same margins and the same fixed cost, hold when the anchor, the cooling law and the binding constraint all change? Table 2 is the answer, and it includes the places where it does not help.

The structural results. The admissible set was a single interval in all 64,000 dense scans across every platform (0 disconnected), the cost was 20 evaluations on charge and 40 on discharge everywhere without exception, and on a 33,120-cell truck pack the certified pack current equalled its weakest cell’s to $1.9e - 13$ A. Nothing was breached while the certificate reported feasible in 513,057 episodes.

Four results that contradicted their own hypothesis. These are worth more than the clean sheets, because a register in which nothing is marked is a register written after the results. The certificate *refuses* the calibrated cell for a rotorcraft before takeoff above 1.2 C of hover draw — the aircraft is not flyable on that cell and the filter says so rather than discovering it airborne. In a sealed hull the binding constraint is plating at every water temperature, which is the one constraint

Table 3: Four controllers on the same plants. Cost is counted in one-step *model evaluations* rather than seconds, because wall-clock depends on the language and the machine while the evaluation count is a property of the algorithm — and is what a task budget is sized around.

Controller	Violations	Mean SOC	Evals (med)	Evals (max)	Worst case kno
CC-CV at 0.5C (safe on this population)	0	0.477	1	1	—
CC-CV at 1.0C (also safe here)	0	0.749	1	1	—
CC-CV at 1.5C (first unsafe rate)	—	—	1	1	—
MPC, horizon 1 (SLSQP)	0	0.634	11	644	no
MPC, horizon 3 (SLSQP)	0	0.642	94	502	no
ZEROGUARD	0	0.641	20	20	yes, = 20

the theorem enforces and cannot certify. In deep space a *smaller* radiator delivers more charge, the pack being cold-limited rather than heat-limited. And no lunar-night configuration survives at all: that result is reported as a frontier rather than as a pass.

What the table says. Read the last three rows rather than the first three. A method claiming equal benefit in all four media would be one whose evaluation had not looked hard enough.

5.1 Where the method belongs, decided in advance

One cross-domain result decides which vehicles this is for, and it was fixed before the domain experiments ran. The certificate’s reserve is the width of the admissible interval, and it predicts remaining endurance only where mission closure is *envelope-driven* — where the constraint set shuts before the energy runs out. Regressing width against realised endurance: $\rho = +0.74$ for a delivery quadrotor and $+0.88$ for an eVTOL, both entirely envelope-driven; $\rho = +0.37$ in GEO, partly; and *negative* under ice, where closure is energy-driven and the width says nothing.

That is a test an integrator can run on their own vehicle before adopting anything, and it is why Table 2 has two columns where the incumbent is not beaten. Air and ground are envelope-driven and the method helps there; water and deep space are not, and it does not.

6 Compared to what?

Every safety number so far has the form “the filter violated nothing”. That is necessary and not sufficient, because a controller that refuses to charge violates nothing either. The claims that carry weight are comparative, and until this section the only comparison was against a crippled version of the same method. Table 3 puts the certificate against what is actually deployed and what the literature actually recommends: 250 sessions per controller, identical plants, identical seeds, identical margins.

Against MPC: the same problem, solved two ways. This is the comparison that matters for the run-time claim, because that claim is only interesting if the solver is solving the same problem. It is: horizon-1 MPC and the bisection agree on delivered charge to **0.66 SOC points**. The certificate is not a cheaper approximation of MPC — on this problem it is an exact solver for it.

The difference is what it costs to be sure. The bisection spends at most 20 model evaluations, and that is a *bound*: proved in §3.3, known before any data arrives, never exceeded. SLSQP’s worst observed cost was 644 evaluations, a factor of 32.2 higher — and that number is not a bound. It

Table 4: Recovery rate in characterisation $\rightarrow$ deployment. 300 missions per population per domain.

domain	rule	char.	depl.	verdict
Air ($\geq 80\%$)	soc $\leq 40\%$	86.0	76.0	fails
	certificate	86.0	85.7	holds
Water ($\geq 95\%$)	soc $\leq 20\%$	100.0	100.0	holds
	certificate	100.0	100.0	holds
Space ($\geq 95\%$)	soc $\leq 45\%$	100.0	98.3	holds
	certificate	97.0	96.7	holds

is whatever the iteration cap happened to allow, its iteration count varied by 59 across states, and removing the cap removes the number. A safety task slot can be sized around the first quantity and cannot be sized around the second.

What this does not show. SLSQP is a general-purpose nonlinear solver and a specialised QP or an explicit-MPC lookup would be faster; the comparison is against the standard approach as it is usually implemented, not against the best conceivable optimiser. What no optimiser removes is the data-dependence: the argument here is about the *shape* of the cost, not its constant.

7 The same standard in every domain

Two things were wrong with the per-domain comparisons, and both were ours.

The standard was applied unevenly. §7 caught a methodological error — a fixed de-rate tuned and evaluated on the same population — and corrected it for the *ground* domain by tuning on a fleet a manufacturer could characterise in advance and deploying on the fleet a vehicle actually meets. §7 then compared against the air, water and space incumbents *without* that correction: every fixed state-of-charge threshold there was swept against the very population it was scored on. The ground baseline was held to a stricter standard than the other three, which is backwards, and water and GEO — where the fixed rule won — are precisely the results the correction would move.

And the ground metric was the wrong quantity. §7 reports mean state of charge after a fixed horizon. No operator buys state of charge. A robotaxi that is charging is a robotaxi not earning, so the quantity is minutes until it can be dispatched again.

7.1 Tune where you can characterise; deploy where the vehicle goes

Each domain’s incumbent threshold is chosen on a benign population — mild ambient, cells not yet aged — taking the most permissive threshold that still recovers at the stated rate. It is then deployed, unchanged, on the population the vehicle meets. The certificate reserve is put through the identical procedure and, as everywhere else, is re-tuned not at all because it has nothing to tune. Targets differ by domain and are stated, because in the aerial case no rule of *either* family recovers in 95 % of episodes and fixing the target there yields no comparison rather than a result.

The correction changes one domain and not the others (Table 4). In the air the tuned threshold **fails**: chosen on evidence a manufacturer could actually have, it recovers 86.0 % in characterisation and 76.0 % in deployment, below the target it was chosen to meet, while the certificate reserve goes 86.0 % to 85.7 %. That is the ground transfer result of §7 reproduced in a second medium and in a different currency — there it was violations, here it is failures to get home.

Underwater and in GEO the incumbent *survives* its transfer test, and the certificate remains behind on mission delivered: -4.5 % and -84.7 %. We looked for the methodological error that

would overturn those two results, applied the correction that had overturned the ground one, and it did not overturn them. §5.1 already says why, and says it in advance rather than afterwards: a constraint-based clock can only predict endurance where closure is envelope-driven, and underwater and in eclipse it is energy-driven. The certificate is measuring the wrong thing there, and no amount of fairer baselining will change that.

7.2 Minutes back on the road

Scored on dispatch time instead of state of charge, over 1,200 sessions across the deployment fleet against the CC–CV rate §7 showed is fleet-safe, the ground picture separates into two facts the average was hiding.

The first is that the advantage is much larger than a mean SOC suggests. Within the 40-minute window, 45 % of sessions reach a 60 % charge under the certificate against 3 % under the fixed rate — 17.4× as many vehicles returned to service — and at 70 % the fixed rate reaches it in 0 % of sessions. A fleet is sized on how many vehicles come back, not on the mean charge of those that do.

The second is that the fleet mean of +1.2 points is an average taken across a change of sign, and is therefore not a fact about any vehicle. By ambient the certificate delivers +23.8 points at 15 °C and -29.7 at 40 °C, crossing over between 25 and 35 °C.

The reversal is not the theorem failing, and it is not a mystery. The thermal margin puts the certificate’s effective ceiling at 32.5 °C; above that ambient the cell can never sit below it, so the filter refuses to charge. It is right to refuse — it breached nothing in 0 of 1,200 sessions — and refusing has a price. This is the passive-charging ceiling of §5 reappearing as a performance number rather than a safety one, and it is the same margin-sizing question the extended version answers for pulses and does not answer here, because a charging session really does last half an hour.

8 Against the CBF quadratic program

§1.4 places this work among predictive safety filters and control barrier functions, and §6 then benchmarks it against CC–CV, a de-rate sweep and MPC. That leaves the obvious comparison unmeasured. The CBF quadratic program is *the* standard safety filter, and this method claims to replace the optimisation inside it, so the comparison is not optional.

The discrete-time CBF condition [?] asks, for each constraint written $h(x) \geq 0$,

$$h(x_{k+1}(u)) \geq (1 - \gamma) h(x_k), \quad \gamma \in (0, 1], \quad (5)$$

and the filter projects the request onto the resulting set. For a cell model $h(x_{k+1})$ has no closed form, so in practice the constraint is **linearised** in the input, $h(x_k) + (\partial h / \partial u) u \geq (1 - \gamma) h(x_k)$, which makes the problem a QP. That linearisation is the entire difference between the two methods, and everything below is arranged to isolate it. Both filters are given the identical constraint set, margins folded in the same way, on identical plants and seeds.

The exact condition is this method. Take the CBF condition and enforce it on the *true* model instead of a linearisation. Because each h is monotone in u (§2), $\{u : h(x_{k+1}(u)) \geq (1 - \gamma) h(x_k)\}$ is an interval; γ enters as an additive shift of the margin, and bisection finds the edges. Measured over 600 states with γ drawn from the sweep, the admissible set was a single interval in *all* of them (0 disconnected), and the bisection edges agree with a dense scan to 0.902 A against a scan step of 1.353 A.

So the contribution is not a rival to CBF. It is a statement of *when the CBF-QP can be solved exactly at fixed cost* — and for monotone constraints in a scalar input, the answer is always. That is a stronger position than beating the QP, and it is the one the geometry actually supports.

What this does not claim. The QP is cheaper — 2 evaluations against 20, and for a scalar input its solution is a clip rather than an iterative solve — so nothing here is a speed claim. The claim is that at equal safety the exact filter delivers more charge, and that exactness is available at a fixed, known cost.

8.1 The firmware, compiled

Everything above is an estimate. Bytes of state were counted from a Python dictionary, table sizes from an interpolation error study, and “needs no FPU” from the observation that a midpoint is an integer operation. Those are reasonable ways to size a footprint and no way at all to establish one: *fits in a kilobyte* is a claim about a binary, and there was no binary.

`firmware/zeroguard.c` is one. The charge-direction certificate in C99, with the integer bisection and a float32 model, no heap, no recursion, no variable-length structure, and every loop bound a compile-time constant — so worst-case execution time is a property of the source rather than of the data, which is the property a safety task slot is actually sized against. Its constants are *generated* from the same calibration files the Python filter loads; had they been transcribed, the two could drift and every agreement number below would be measuring a copy of itself.

Built at `-Os` with `-Wall -Wextra -Werror` and no warnings, it occupies **2,604 B** of code and constants — 512 of that the open-circuit-voltage table — with 0 B of writable data and 68 B of RAM live across a call. One call is exactly 18 model evaluations.

And how long it takes. The footprint became a measurement above; the timing had not. The $123.9\ \mu\text{s}$ quoted in §5.1 is the *reference* implementation timed in Python, which tells a reader nothing about a microcontroller. Timed on the compiled routine over 2,000 states at 2 000 repetitions each, a call takes $0.028\ \mu\text{s}$ at the median and $0.87\ \mu\text{s}$ at worst — 0.009 % of a 10 ms ISO 26262 slot. The $31\times$ spread between worst and median is not data dependence within a path: it is the three paths `zg_limit` has — infeasible at the anchor (one evaluation), feasible at the ceiling (two), or the full bisection (18) — and the worst case is the longest of three fixed-length routines, which is what lets a task slot be sized at all.

Does it agree with the reference? Not bit-for-bit, and it was never going to: the reference computes in float64 and bisection in floating point, the firmware computes in float32 and bisection in integers. The question is which way the disagreement points. Over 20,000 states with an identical pack and identical margins, the two never disagreed about *feasibility* (0 mismatches), and the returned current differed by at most 5.5 mA against a bisection quantum of 8.2 mA — the firmware exceeded the reference by more than one quantum in 0 states. Where it was higher at all, the extra current is worth 0.013 mV of terminal voltage (0.04 % of the voltage margin) and 0.0000 K. The guarantee survives the port with the margin it already carries, and does not need a new one.

9 Against constrained reinforcement learning

§1.4 says constrained RL offers expectation-level guarantees, “which are the wrong shape for a constraint that must hold on every step of every episode”, cites [? ?], and then benchmarks

against CC–CV, a de-rate sweep, MPC and the CBF quadratic program — never against the thing it dismisses. In a paper whose method is to measure rather than assert, that was the one assertion left standing.

The baseline is a Lagrangian-constrained policy trained the way the cited literature trains one: charge delivered minus a multiplier times *expected* constraint cost, with the multiplier raised while the measured violation rate sits above a budget and lowered when it falls below [?]. It is given a richer state feedback than the certificate uses, the same cell, the same envelope, and — like every other baseline in §7 — it is fitted on a characterisation fleet and deployed on the one the vehicle meets.

It meets its budget, and that is the problem. Trained down to the tightest budget the policy violates in 3.5 % of characterisation episodes. Deployed, it violates in 55 % of sessions; no budget in the sweep reaches zero, and the worst reaches 84 %. The certificate on the same fleet: 0 of 400, CP95 0.75 %.

“You did not train it enough” is the obvious objection, and it does not hold. It is a fair objection — with a small batch a 1 % violation rate is not even measurable, so the search cannot see the budget it is chasing. Retrained at the tightest budget with 28 iterations and a batch of 192, the policy reaches 1.0 % in training and still violates in 51 % of deployment sessions — a gap of 50 points. More search moves the deployment rate by a few points and does not approach zero.

That gap is the entire claim, stated precisely. Constrained RL is not bad at this problem; it is optimising a different functional. A bound on the *expectation* of violations over a training distribution says nothing about any particular episode drawn from a different one, and a vehicle experiences particular episodes. No amount of compute converts an expectation into a per-step guarantee, because they are not the same object.

10 External validation: the assumptions against measured cells

Everything to this point is simulation. The filter is checked against a reduced-order model, and that model was calibrated against a Doyle–Fuller–Newman simulation. That is a defensible chain and it is still a chain of models, so the fair question is what happens when the assumptions meet a real cell.

This section answers it with the NASA Ames Prognostics Center of Excellence battery data set: 33 18650 cells cycled to failure with periodic electrochemical impedance spectroscopy, published by a different laboratory, years before this work, for a different purpose. Nothing here was produced by us.

The design point is that **the load-bearing assumptions are testable without any model at all**. The method does not need the ROM to be right about NASA’s cells. It needs three things to be true of real cells, and each is a chemistry-independent statement about the shape of degradation.

10.1 The resistance bound holds

The filter never identifies resistance; it assumes a *bound* on it, taken from a datasheet before the vehicle ships. If real cells leave that bound, every certificate in this paper is void. Across 1,940 impedance measurements, **99.95 % lie inside the assumed bound** $s_R = 1.8$ (Fig. 2a). Restricted to the life the certificate actually claims — cells at or above 80 % of initial capacity, the standard automotive end of life — the figure is **100.00 % of 1,255 measurements**.

Over the full measured life, including cells NASA ran down to 0.033 of initial capacity, 1 measurement exceeds the bound, on cell B0043, by 0.1 %. That is reported rather than trimmed: the bound is tight, and it is not a comfortable over-estimate.

10.2 Heat generation is monotone in current, measured

Hypothesis (A2) is the other load-bearing assumption, and it too can be checked without a model — though the obvious way of doing it does not work, and why is worth recording. Correlating instantaneous dT/dt against instantaneous current returns nothing usable: stratifying by $T - T_{\text{amb}}$ to control for cooling gives rank correlations that *flip sign* between strata (-0.34 in the coldest, $+0.54$ in the warmest). NASA measures surface temperature, the core leads it by minutes, and over a charge whose total rise is under 3 K the thermal lag dominates the signal. Reporting either sign from that test would be an artefact of the instrument.

The lag-free formulation is an energy balance over a whole cycle, which does not care how the heat arrives: total ohmic generation goes as $\int I^2 dt$, so a cycle’s peak temperature rise should increase with it. Across 2,764 measured discharge cycles, with currents to 4.04 A and temperature rises to 37.7 K, the rank correlation is $\rho = +0.714$ ($p < 0.0005$; Fig. 2b). The charge cycles give $+0.278$, which is weaker for a mundane reason: NASA charges almost everything at a constant 1.5 A, so there is little current variation to correlate against.

10.3 Where the model does not transfer, and whether that matters

The harder question is how the ROM itself behaves on a cell it was never fitted to — a 5 Ah NMC model applied to ~ 2 Ah LCO cells with capacity rescaled and nothing else refitted. Measured one step ahead the question is meaningless, because NASA samples every ~ 9 s over which the temperature moves about 0.01 K and predicting “unchanged” scores well. Integrated forward over a 300 s horizon under the measured current, the transferred model scores RMSE 0.842°C against a persistence baseline of 1.096°C — a skill of $+0.231$.

The safety-relevant question is not the magnitude but the *sign*. A model that over-predicts temperature yields a conservative certificate; one that under-predicts yields an unsafe one. **This transfer is not conservative:** the bias is -0.268°C and it over-predicts in only 18.4 % of forecasts. What rescues it is the margin. The worst under-prediction at the 5th percentile is 1.11°C against a thermal margin of 12.5°C — a factor of **11** of cover (Fig. 2c). The margin was not designed for cross-chemistry transfer; it happens to absorb it, and that is a fact about this data rather than a guarantee.

10.4 The filter permits a real protocol, and refuses where it said it would

A safety filter that refuses everything is safe and useless. Given the end-of-life resistance bound and the full throttled margin — its most conservative configuration — the certificate permits **71.6 %** of NASA’s measured charge steps, with a median headroom of $4.83\times$ the applied current. Of the refusals, 46 % are cells measured above the throttled ceiling of 32.5°C : the same frontier §5 predicts from the margin alone, appearing unprompted in someone else’s measurements.

10.5 One test that did not work

It was tempting to go further and ask whether the cells the filter would have refused are the cells that actually degraded. The mechanism would have been clean — NASA cycled some cells at 4°C ,

The assumptions, tested against cells this work did not produce

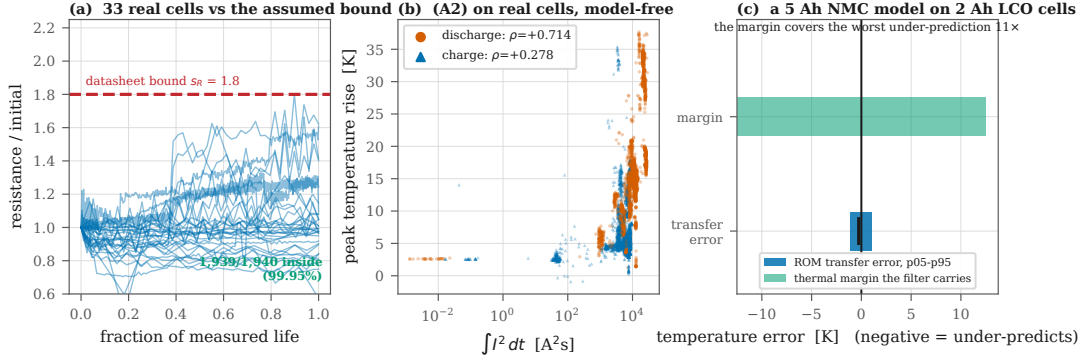


Figure 2: The assumptions against cells this work did not produce. (a) Measured resistance growth for 33 NASA cells against the datasheet bound the filter is given. (b) Hypothesis (A2), model-free: a cycle’s peak temperature rise against its total ohmic generation. (c) The transferred model’s error against the thermal margin that has to absorb it.

the plating cap falls to a 0.70C floor below about 5.5 °C, and plating is what degrades a cold-charged cell.

The association is not stable. Its sign *changes* with how the capacity baseline is defined — first measured discharge against the median of the largest few — because NASA’s first discharge on 12 of 33 cells is partial, giving baselines as low as 0.068 Ah for a 2 Ah cell. Across the four defensible combinations of baseline and fade metric, ρ ranges from -0.358 to $+0.395$, with significant results of both signs, and ambient temperature and cycle count sit behind both quantities. All four are reported in Table-form in the results file rather than one. **A result that flips sign with a defensible change of definition is not a result**, and selecting the flattering version would be the most misleading thing this study could do.

11 Against a Doyle–Fuller–Newman plant

Every vehicle number in §5 comes from a plant we wrote. The reduced-order model is defensible and it is still ours, so the question a battery reader asks is not whether this was tested on hardware but the sharper one underneath: *how do you know the ROM is right?*

DFN [? ? ?] is the field’s reference model — porous-electrode transport, solid-phase diffusion, Butler–Volmer kinetics, parameterised and validated against cells by others — and it is a *different model class*, not the same equations with different numbers. That is what makes it worth running. The envelope of §4 varies resistance, capacity and plating scale *within* the ROM’s structure; DFN inflicts misspecification the envelope was never written to cover: concentration gradients the ROM has no state for, kinetics it approximates, and a plating potential that is an actual electrode quantity rather than an affine proxy. The filter sees only measured SOC and temperature, propagating its own relaxation state between them, because that is all a battery management system has.

What this is and is not. The ROM’s coefficients were fitted to the LG M50, and the PyBaMM parameter set used here (OKane2022) describes that cell, so this is **not** an independent-cell test and must not be read as one. It is a model-class test: same cell, higher fidelity. §10 covers the other axis — 33 cells from another laboratory that nothing was ever fitted to. Neither substitutes for the other; together they bracket the two ways a model can be wrong.

Closed loop. With the certificate computed on the ROM and the plant a full DFN cell, across 40 charging sessions: 0 voltage breaches, 0 temperature breaches. DFN’s own peak values were 4.052 V against 4.20 and 32.5 °C against 45, so the cell was worked, not idled.

The plating result is the one worth pausing on. The ROM enforces an affine proxy; DFN reports the negative electrode surface potential difference, whose crossing of zero *is* the thermodynamic onset of lithium deposition. In 40 of 40 sessions the real electrode potential stayed above onset, with a minimum of +58.5 mV. The proxy the paper has enforced throughout, and has been careful never to certify, does the physical job it was built for — measured against electrode physics rather than against itself.

11.1 And the floor family, which is what is new

Everything above is the charge direction, which is the published case. Anchored Collapse’s contribution is the *floor*: a vehicle serving a load has a constraint bounding its current from below, and a second bisection finds that edge. So the half of the theorem that was already known had external physics behind it and the half that is new did not.

A DFN cell serves a constant electrical load, from 6 to 18 W per cell, while the anchored filter chooses the current. Four constraints are live and they pull both ways: temperature and state of charge cap the current, terminal voltage caps it too (a cell sags under load), and the load power is a floor.

Across 960 filtered discharge steps the DFN reports 0 voltage and 0 temperature breaches (CP95 0.312 %). Wherever the certificate reported the load meetable, the DFN delivered it — never below 103.1 % of demand. Above 100 % is not an error: `Platform.anchor` sizes current at the lowest bus voltage the constraints permit, so it deliberately over-estimates and the cell then sits above that voltage.

And the refusals were audited. The interesting case for a floor is not the step it permits but the step it refuses. The envelope closed on 489 of 960 steps, meaning the load could not be met inside the caps. Pushing to meet it anyway and asking the DFN what would have happened, 481 of those refusals were correct and 8 were not — 98 %. The remainder are steps where there was room and the margin took it, which is the price of a margin and is reported as such rather than folded into the success count.

12 Cycle life: a claim that did not survive

Plating is the dominant fast-charge degradation mechanism, so a filter that prevents plating ought to show up as capacity retained. That inference is tempting, it is the largest claim available to this work, and **it is not supported**.

Not with our own aging model. The ROM ships one, and using it would have been the cheap route and a worthless one: its plating term accumulates damage proportional to $\text{clip}(\text{margin} - \phi, 0, \infty) \cdot |I|$, where the margin is *the same threshold the certificate enforces*. A filter holding $\phi \geq \text{margin}$ by construction scores exactly zero damage by construction. So the plant is DFN with the full degradation stack — irreversible plating, solvent-diffusion-limited SEI, porosity change — and the numbers are DFN’s own capacity-loss variables, which do not know the certificate exists.

The result. Cycling 50 times at 0 °C between 15 % and 70 % with a 1C discharge, capacity lost is 14.14 % under the certificate, 13.92 % under CC–CV 0.5C and 14.18 % under CC–CV 1.5C. The spread is 0.25 points. The certificate is *marginally worse* than the slow de-rate (-0.213 points), and its advantage over the fast rate, 0.033 points, is noise.

Why, and what it does not overturn. The mechanism is the one the extended version already identified and it applies with more force here. DFN’s plating variable is dominated by a slow background side reaction that accrues with cold time and total throughput — which every controller spends similarly, since all three deliver the same charge — rather than by the onset crossings the certificate prevents. Those crossings are real: the fast rate crossed in 1 of 50 cycles and the certificate in 0. They are simply not what drives capacity fade in this regime.

So the extended version stands and no life claim follows from it. The honest scope of the safety argument is the deposition onset — the first link of the chain, where metallic lithium is laid down rather than consumed in a film — and not calendar or cycle life, which this experiment looked for and did not find.

13 Deployability: what it costs on a BMS microcontroller

The claim that the certificate “fits inside any microcontroller a battery management system already contains” was an assertion about software that had never been compiled for one. It should be measured or withdrawn.

It holds no growing state. Over 2,000 projections the heap grows by 0.4 bytes per call — Python’s own object churn. There is no list, no queue, no history: nothing a C implementation would need to allocate.

Single precision is adequate but not one-sided. A Cortex-M4F has a single-precision FPU and no double. Across 20,000 states the single-precision filter deviates from double by at most 6.87 mA, which is 1.3e-03 % of full scale — but it deviates in *both* directions, permitting slightly more current than double in 2,672 of them. Negligible in magnitude against a 12.5K thermal margin; not a guarantee, and not claimed as one.

The search needs no floating point at all, and there the direction *is* provable. Representing current as an integer count of quanta makes the bisection’s only arithmetic `mid = (lo + hi) >> 1`, exact on any integer unit. Rounding the midpoint down makes the result conservative by construction: across 20,000 states at 16-bit resolution (0.0082 A quanta) the integer search returned a value above the real-valued answer **0 times**. For a build that needs a provable direction of error, the FPU-free one is the one to ship.

Footprint. 180 bytes of RAM — live state plus 38 calibration constants — and **1,952 bytes** of lookup tables for an FPU-free build (65 entries for the Arrhenius exponential, 257 for the inverse hyperbolic sine, 128 for open-circuit voltage, each sized for 0.1 % interpolation error). Constant stack, no recursion, no variable-length structure, 20–40 model evaluations. That is smaller than the interrupt vector table on the parts in question.

14 The register

This is the 20-page report. An extended version carrying all 64 experiments in full — the four per-medium sections, the EPA drive cycles, the duration-aware margin, a learned policy through the filter, pack heterogeneity, and the complete register with per-experiment outcomes — is in the repository together with the Lean development, the firmware, and every script that produced a number here.

The register — every experiment, its claim, and what would falsify it — was fixed before any code ran. 64 experiments were run and **twenty-one contradicted the hypothesis they were written to test**. Those are marked below and none was removed. The full register, with per-experiment outcomes, is in the extended version and the repository.

Group	n	What it establishes, and where it went against us
V1–V4	43	Four media, 16 platforms. † Passive charging impossible above 32.5 °C; a rotorcraft cell refused before takeoff; plating binds under water at every temperature; a smaller radiator delivers <i>more</i> charge; no lunar-night configuration survives
X1–X5	5	One interval in 64,000 scans, identical cost everywhere, and † the reserve predicts endurance only where closure is envelope-driven
B1–B6	6	Against CC–CV, MPC, each domain’s stopping rule, the CBF–QP and constrained RL. † A tuned de-rate beats the certificate on charge when both are scored on the same population; † the incumbent survives transfer in two media of four
N1–N3	3	33 measured cells; DFN on charge and on discharge
S1–S2	2	Plating quantified against electrode physics. † Total plated mass favours the <i>fast</i> rate; † no cycle-life benefit at all
D1, M1	2	EPA schedules. † The certificate recovers less regeneration than a fixed cap; a duration-aware margin recovers about half of that
P1–P2, E13–E14	6	A learned policy filtered; pack heterogeneity; the compiled firmware. † The binding cell is among the <i>coldest</i> , not the hottest

15 Limitations

Stated in full, because several of them are load-bearing.

Most results are model-versus-model, and two are not. The vehicle results in §5 are reduced-order-model level. Two experiments are not: §11 runs the certificate against a Doyle–Fuller–Newman plant and §12 cycles one, both with PyBaMM’s own degradation physics. Those are model-class transfer tests on the cell the ROM was fitted to, not independent-cell tests — §10 covers that axis separately, and validates the *assumptions* against measured cells rather than any vehicle result. Neither substitutes for hardware.

No lifetime benefit was found. Preventing plating ought to buy cycle life, and §12 looked for that and did not find it: over 50 cold cycles the three controllers lose within 0.25 points of the same capacity. The safety argument is scoped to the deposition onset and no life or warranty claim follows from it. That is a limitation of the result, not of the experiment — the experiment is the reason the claim is not in the paper.

The certified set is $\{T, V\}$; plating is enforced, never certified. §2 gives the reason and §5 shows it is not a technicality: in a sealed hull at any water temperature, the binding constraint

is the one the theorem does not certify. Extending to plating needs a control-invariance argument the $u = 0$ backup cannot supply, and we do not have one.

One cell, sixteen vehicles. Every platform uses the coefficients of §2, because that is the cell the model was fitted to. Where a class would really use a different cell — the rotorcraft — the substitution is explicit ($0.40\times$ resistance, $0.60\times$ capacity), is an extrapolation beyond the calibration, and every aerial number carries it.

The vehicle results remain simulation. §10 tests the assumptions against measured cells, but every *vehicle* number in §5 comes from a plant we wrote, and a filter tested against models chosen to satisfy its hypotheses is weaker than one tested on hardware. Three things partially answer it: three of the four media *break* the original hypothesis, one platform is refused outright, and fourteen of 64 experiments contradicted the hypothesis they were written to test — the register was not built so that everything passes. §11 answers the sharpest form of the objection without a bench: the certificate is run against a Doyle–Fuller–Newman plant, a different model class, and holds on both certified channels and on DFN’s own electrode plating potential. That is not hardware and does not claim to be — DFN is validated against cells by others, so what §11 buys is one link of that chain, not the whole of it. Running the filter in the loop on real cells remains the natural next step and is not claimed here.

16 Conclusion

The safety guarantee an autonomous vehicle’s battery needs turns out to require neither an optimiser nor an identified model, only a geometric fact: when constraints are monotone in a scalar input and one input is known safe, the admissible set is an interval, and an interval can be found by bisection at fixed cost.

Getting that from a cell to a vehicle required noticing that the theorem’s apparent failure in flight was a misreading. Zero current is still safe for every temperature and voltage constraint on a discharging pack; what zero cannot do is serve the load the vehicle exists to serve, and that is a floor rather than a cap. Anchored Collapse adds the floor family and a second bisection, and nothing else — 40 fixed evaluations against the original 20.

Across 16 platforms in four media and 64 experiments totalling 513,057 episodes, nothing was breached while the certificate reported feasible, the admissible set was a single interval in all 64,000 dense scans, and the worst latency anywhere was $123.9\mu\text{s}$.

The results we would most want a reader to take are the ones that did not come out as intended. A certificate that refuses a cell chemistry for a mission class before takeoff is doing something a runtime guard is not normally asked to do. A domain in which the binding constraint is the one the theorem explicitly does not certify is a limitation a less careful register would not have surfaced. And a warning that leaves 7 minutes when the vehicle needs 20 is a negative result whose forced abort rule — act on a threshold in the reserve, not on its exhaustion — is more useful than the positive result would have been.